

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	8 March 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The system collects EV charging and driving datasets, processes the data, performs analysis, and visualizes insights using an interactive dashboard.

Architecture Flow:

Data Source → Data Processing (Python) → Database Storage → Tableau Visualization → Web Dashboard

S.No	Component	Description	Technology
1.	User Interface	Dashboard where users view EV charging and range insights	Tableau Dashboard / Web UI
2.	Application Logic-1	Data preprocessing and analysis of EV charging dataset	Python (Pandas, NumPy)
3.	Application Logic-2	Exploratory data analysis and pattern detection	Python (Matplotlib, Seaborn)
4.	Application Logic-3	Data transformation for visualization	Python Scripts
5.	Database	Stores EV charging data such as battery capacity, charging time, energy consumption	MySQL / CSV Dataset

6.	Cloud Database	Optional cloud storage for dataset and analysis	Google Drive / Cloud Storage
7.	File Storage	Storage of dataset files and processed data	Local File System
8.	External API-1	EV charging station information (if required)	Open Charge Map API
9.	External API-2	Mapping and location visualization	Google Maps API
10.	Machine Learning Model	Predict EV range based on battery and energy consumption	Scikit-learn (Regression Model)
11.	Infrastructure (Server / Cloud)	Application execution and visualization deployment	Local System / Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Tools used for data analysis and visualization	Python, Pandas, NumPy, Matplotlib
2.	Security Implementations	Secure data access and safe data storage	User authentication, Data encryption
3.	Scalable Architecture	The system can handle larger EV datasets and more visualizations	Modular data processing architecture
4.	Availability	Dashboard accessible anytime through Tableau	Tableau Public / Web Hosting
5.	Performance	Fast data processing and visualization for EV analytics	Optimized python scripts and Tableau dashboards

